

Decoding Longitudinal Affective Dynamics through Specialized Transformer Architectures

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Abstract

This paper investigates LLMs’ ability to quantify human emotions through text and temporal sequences within the SemEval task on valence and arousal variation. We evaluate model performance in predicting affective scores for isolated sentences and emotional shifts across longitudinal windows of comments. By comparing three RoBERTa variants pre-trained on diverse datasets, we examine how domain-specific knowledge influences the perception of subjective affect. This study moves beyond static snapshots to determine the extent to which Transformer-based architectures can capture dynamic emotional trajectories in natural environments.

1 Introduction

Human emotions are not static states but dynamic, subjective processes that evolve over time in response to external stimuli and environmental contexts. Traditionally, Natural Language Processing (NLP) has treated emotion recognition as an isolated classification problem, relying on static snapshots and third-party annotations. However, human affect is inherently longitudinal and deeply rooted in an individual’s internal experience.

1.1 The SemEval Framework and Ecological Dataset

In this study, we address the challenge proposed by the **SemEval-2026 Task 2** (1), which marks a fundamental paradigm shift in affective computing. Unlike conventional datasets based on public social media comments, the data provided for this task consists of *longitudinal ecological essays and feeling words*. These texts offer a unique introspective perspective, as they were collected directly from participants in their natural environments over a multi-year span (2021–2024). This longitudinal nature allows for the analysis of not only momentary

affective states but also the emotional trajectories and individual rhythms of the subjects.

1.2 Architectural Backbones

To investigate the ability of Transformer-based models to interpret these complex affective states, we selected three variants of the **RoBERTa** architecture (2), each characterized by a distinct specialization derived from its pre-training process:

- **RoBERTa-base:** A robust evolution of BERT, pre-trained on a massive 160GB general-purpose corpus. In our study, it serves as the foundational linguistic baseline.
- **Twitter-RoBERTa-Emotion:** Developed by CardiffNLP (3), this variant underwent domain adaptation on 58 million tweets. It is specifically optimized to capture the informal expressiveness and linguistic nuances typical of social-media-style entries.
- **RoBERTa-GoEmotions:** Fine-tuned on the GoEmotions dataset (4), this model leverages 50,000 Reddit comments labeled with 28 distinct emotional categories. It represents a state-of-the-art approach for mapping fine-grained emotional taxonomies into continuous dimensions.

1.3 Contributions

This study provides a comprehensive analysis of longitudinal affect modeling with three primary contributions: (1) **Architectural Specialization:** We identify a performance dichotomy where **RoBERTa-GoEmotions** excels in static assessment (Subtask 1), while **Twitter-RoBERTa** and **Mental-RoBERTa** are superior for forecasting state changes (2A) and dispositional drift (2B), respectively. (2) **Personalization:** We demonstrate that learnable **User Embeddings** are essential to model individual "affective baselines" in ecological

data. (3) **Serialization Strategy:** We introduce a context-aware methodology that encodes tabular metadata and historical scores directly into textual inputs, enabling standard Transformers to capture temporal emotional momentum without architectural overhead.

2 Related works

Dimensional Affect & Transformers. Our work builds on Russell’s (5) circumplex model, mapping emotions onto continuous Valence-Arousal dimensions rather than discrete categories. This is implemented via pre-trained Transformer architectures, specifically leveraging the fine-grained taxonomies of the GoEmotions dataset (4) adapted for continuous regression tasks.

Personalization & Robustness. To address the subjectivity of self-reports, we employ latent User Embeddings (6) to learn individual "affective baselines" and calibrate regression outputs. To mitigate noise and outliers inherent in longitudinal ecological data, we transition from standard MSE to robust loss functions like Huber Loss (Smooth L1) (7), ensuring stability against unrepresentative data points.

Longitudinal Modeling & Domain Adaptation. Beyond static analysis, we model emotional trajectories using a sliding context window to capture temporal momentum (8). While previous works used RNNs/GRUs for dialogue or mood shifts (9; 10), we utilize Transformer attention over sequences. Furthermore, we leverage domain-adaptive models like *MentalRoBERTa* (11), which provide psychologically-grounded representations superior to general-purpose models for introspective and longitudinal affect tracking.

3 Dataset Analysis

In this section, we analyze the composition and distribution of the dataset used for Subtasks 1 and 2. The data is provided in a structured format containing the following fields: `user_id`, `text_id`, `text`, `timestamp`, `collection_phase`, `is_words`, `valence`, and `arousal`. Our exploratory data analysis highlights key characteristics that informed our subsequent modeling choices.

3.1 Data Composition and Textual Modalities

The dataset consists of two distinct types of textual expressions, identified by the `is_words` flag:

- **Essays/Phrases:** Free-form narrative texts or structured sentences, totaling 1,331 samples.
- **Feeling Word Lists:** Non-syntactic lists of individual words describing emotional states, totaling 1,433 samples.

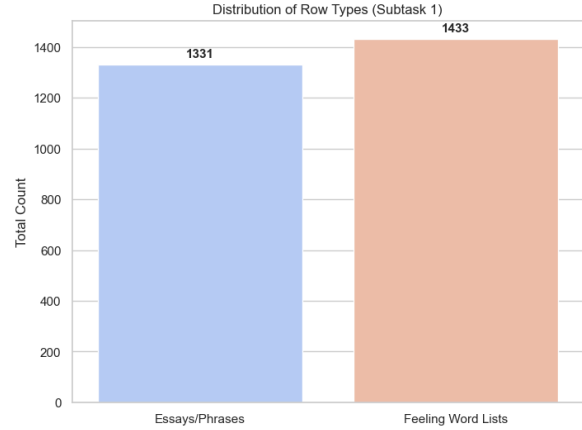


Figure 1: Distribution of textual modalities (Essays vs. Feeling Word Lists) in the Subtask 1 dataset.

This balanced distribution between narrative and list-based inputs requires models capable of processing both complex syntax and isolated semantic cues.

3.2 User and Word Distributions

The longitudinal nature of the tasks necessitates a clear understanding of user contributions and vocabulary sparsity.

- **Texts per User:** As shown in Figure 2, approximately 40% of users provided fewer than 10 texts. This led us to define a *Minimum Comments* (MC) threshold of 10 to distinguish between “known” users for personalized embeddings and generic user profiles.
- **Word Occurrences:** Vocabulary analysis reveals that over 80% of target words appear fewer than 10 times in the corpus (Figure 3). This high level of sparsity justifies the use of *Permutation Augmentation Value* (PAV) for word lists to maximize the learning signal from infrequent terms.

3.3 Affective Space Density

The distribution of samples within the Valence-Arousal circumplex is non-uniform, as illustrated by the density heatmap in Figure 4.

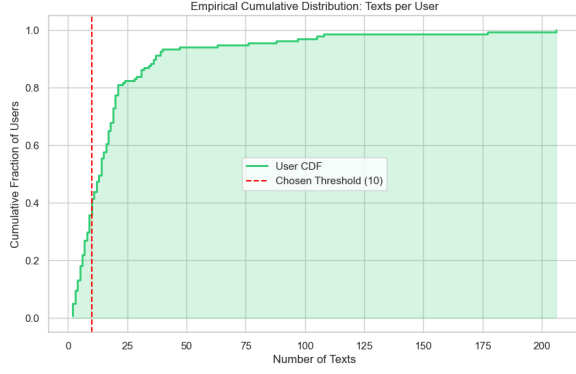


Figure 2: Empirical Cumulative Distribution Function (ECDF) of texts per user.

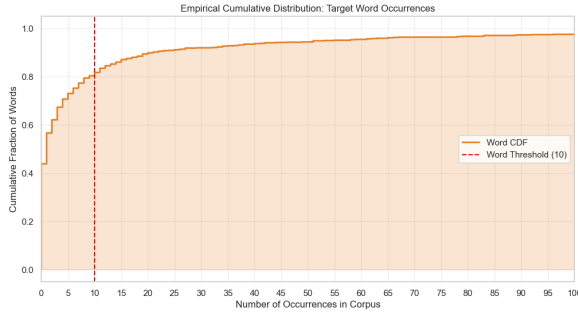


Figure 3: Empirical Cumulative Distribution Function (ECDF) of target word occurrences.

- **Central Tendency:** The highest density of samples (473) is concentrated around the neutral-to-positive center of the space (Valence 0.0, Arousal 1.0).
- **Arousal Distribution:** There is a significant concentration of low-arousal states (0.0), with 361 samples specifically mapped to high valence and low arousal.
- **Data Imbalance:** We observe a lower frequency of samples in extreme quadrants. This imbalance suggests that models may naturally achieve higher precision in predicting neutral or positive states compared to high-intensity emotional peaks.

4 Subtask 1: Longitudinal Affect Assessment

4.0.1 Task Definition

Subtask 1 involves point-wise assessment of affective states via regression. Given a text e_i (essay or word list) from a user, the goal is to predict continuous values $\hat{y}_i = (\hat{v}_i, \hat{a}_i)$, where $\hat{v}_i \in [-2, +2]$ is **Valence** and $\hat{a}_i \in [0, 2]$ is **Arousal**.

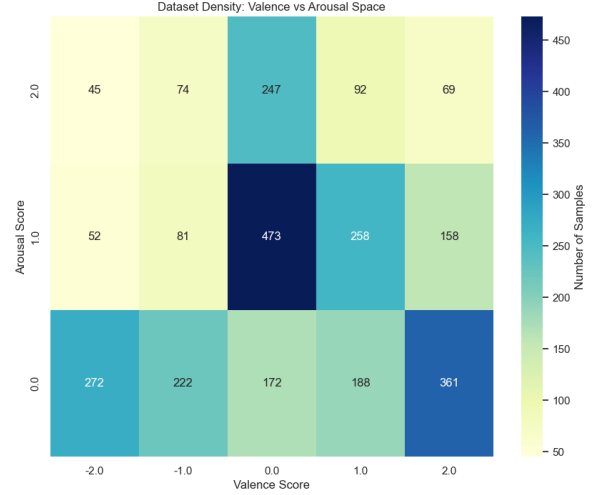


Figure 4: Density heatmap of the dataset across the Valence vs. Arousal space.

4.0.2 Methodology

We optimize the $e_i \rightarrow (\hat{v}_i, \hat{a}_i)$ mapping by integrating domain-specific pre-training and personalized user modeling.

Affective Backbones. We evaluated three RoBERTa variants: *base*, *Twitter* (for informal text), and *GoEmotions*. The latter serves as our primary backbone due to its fine-tuning on complex emotional taxonomies.

User Embeddings (UE). To handle subjectivity, we utilize learnable **User Embeddings**. A vector $\mathbf{z}_u \in R^d$ is concatenated to the Transformer representations to capture each user’s specific emotional range.

Shared Identity Strategy. We manage **Unseen Users** and sparse profiles via a generic identity (ID=0). This provides a robust "average" affective baseline and mitigates noise from low-frequency users.

Data Enhancement (PAV). For word lists, we apply n random permutations (**PAV**) during training to ensure invariance to word order, while preserving the syntactic structure of narrative essays.

Metrics. We use Pearson correlation (r) and *MAE* across three dimensions: **Between-user** (r_{btw}), **Within-user** (r_{wth}), and the primary **Composite** metric (r_{comp}), calculated via Fisher’s z-transformation.

Optimization. The system is optimized using **Huber Loss** to ensure robustness against outliers in self-reports, weighting Valence and Arousal equally.

4.1 Results and Analysis

4.1.1 Experimental Setup

Optimal setup: *max_len* 128, *batch_size* 16, *lr* 2×10^{-5} , *hidden_dim* 128, and *dropout* 0.2. Results are averaged across four random seeds (0, 100, 200, 300).

4.1.2 Comparative Model Performance

As shown in Table 1, RoBERTa models outperform SemEval baselines. **RoBERTa-GoEmotions** is the top performer, achieving an Arousal r of 0.528 and the lowest $MAEs$ (0.662 V, 0.440 A). The +0.229 improvement in Arousal confirms the efficacy of emotional pre-training for navigating the Valence-Arousal space.

Model	Valence (V)		Arousal (A)	
	r	MAE	r	MAE
SemEval Baseline				
linear(BERT)	.557	.743	.299	.459
rand (mean)	.000	.000	.000	.488
Tested Models				
roberta-base	0.671	0.688	0.504	0.458
twitter-roberta-base-emotion	0.686	0.681	0.501	0.456
roberta-base-go_emotions	0.663	0.662	0.528	0.440
Difference with baseline				
roberta-base	+0.114	-0.055	+0.205	-0.001
twitter-roberta-base-emotion	+0.129	-0.062	+0.202	-0.003
roberta-base-go_emotions	+0.106	-0.081	+0.229	-0.019

Table 1: Task 1 Results: Composite metrics analysis for Valence and Arousal.

4.1.3 Impact of Personalization and Robustness

Hyperparameter analysis (Table 2) highlights:

- **UE & MC:** The pairing of User Embeddings (32-dim) and an $MC = 10$ threshold is optimal, improving Valence correlation (+0.0677) and reducing error (−0.0623). The model effectively corrects for individual reporting bias.
- **PAV:** Increasing permutations ($PAV = 3$) without a strong user identity degrades performance, suggesting robustness requires stable user-style representations.
- **Huber Loss:** Stabilized training by reducing sensitivity to self-report outliers while maintaining precision on prevalent states.

Table 2: Valence and Arousal results for RoBERTa-base-GoEmotions with varying hyperparameters.

UE	MC	PAV	Valence (V)		Arousal (A)	
			<i>r</i>	<i>MAE</i>	<i>r</i>	<i>MAE</i>
roberta-base-go_emotions base results						
0	0	0	0.6631	0.6629	0.5283	0.4402
roberta-base-go_emotions mod results						
0	0	3	0.6397	0.6809	0.4958	0.4603
32	0	0	0.6495	0.7155	0.5495	0.4487
32	10	0	0.7308	0.6006	0.5529	0.4140
32	10	3	0.6961	0.6298	0.5052	0.4449
Difference with base results						
0	0	3	−0.0234	+0.0180	−0.0325	+0.0201
32	0	0	−0.0136	+0.0526	+0.0212	+0.0085
32	10	0	+0.0677	−0.0623	+0.0246	−0.0262
32	10	3	+0.0330	−0.0331	−0.0231	+0.0047

4.1.4 Discussion

Gains are significant given the base model’s strength. A +0.06 correlation increase indicates successfully mitigating subjectivity bias. Consistency across Valence and Arousal confirms that UEs provide a genuine affective signal rather than overfitting.

5 SubTask 2: Forecasting Future Variation

Given a sequence of the first t texts (essays or feeling words) together with their corresponding affect scores (either Valence or Arousal), this task asks systems to forecast distinct types of variation based on this history. Let $v_{1:t} = (v_1, v_2, \dots, v_t)$ denote the sequence of observed scores associated with these texts.

5.1 Subtask 2A: State Change

This subtask focuses on the immediate evolution of the emotional state. It requires predicting the change from the last observed texts t to the immediate next text at index $t + 1$. The target variable Δ_1 is defined as:

$$\Delta_1 = v_{t+1} - v_t \quad (1)$$

5.1.1 Forecasting Data Structure

The dataset for Subtask 2 is specifically structured to support longitudinal forecasting, with a clear distinction between training and evaluation phases.

Training Phase. The model utilizes the complete historical timeline of the users. Input features consist of the full sequence of texts (essays or feeling

words), timestamps, and collection phases. Crucially, each historical entry is paired with ground-truth Valence and Arousal scores. The forecasting targets (Δ_1 and Δ_{avg}) are derived by aligning this historical sequence with the user’s subsequent emotional states.

Evaluation Phase and Temporal Boundaries.

The test set introduces a realistic forecasting scenario where future textual content is withheld. Instead, the dataset provides *temporal boundaries* to model time-decay effects:

- **Immediate Future Boundary:** The dataset provides the timestamp and collection phase (`timestamp_min`) corresponding to the immediate next text. This defines the target time for the State Change prediction (Δ_1).
- **Long-term Boundary:** The timestamp of the final sequence text is provided (`timestamp_max`) to define the span for Dispositional Change (Δ_{avg}).

This structure compels models to rely solely on past affect trajectories and time gaps, without access to future linguistic cues.

5.1.2 Methodology

To address the forecasting challenge of Subtask 2A, we formulated the problem as a regression task conditioned on historical context using a Transformer-based architecture.

Input Representation: Sliding Context Window.

We model the user’s emotional trajectory as a sequence of events. Instead of treating each text entry in isolation, we adopted a sliding window approach with a history horizon of $k = 4$. For a target time step t , the input consists of a concatenated sequence of events from $t - k$ to t . This allows the model to capture short-term temporal dependencies and emotional momentum.

Feature Injection via Serialization. To leverage metadata alongside textual content, we employed a text serialization strategy. Tabular features—specifically time (hour (H), month), day of the week (DOW), data collection phase, and the entry type (essay vs. feeling words)—are converted into special textual tokens. Each event i in the history window is formatted as:

$$E_i = [\text{META}_i] \oplus [\text{TYPE}_i] \oplus [\text{SCORES}_i] \oplus \text{TEXT}_i \quad (2)$$

where $[\text{META}_i]$ encodes temporal context (e.g., $[\text{H}=14 \text{ DOW}=2]$), $[\text{TYPE}_i]$ distinguishes between $[\text{ESSAY}]$ and $[\text{FEELING_WORDS}]$, and $[\text{SCORES}_i]$ explicitly includes the valence and arousal values of that past step. This prompts the Transformer to attend to both semantic content and situational context simultaneously.

Target Normalization. The regression targets ($\Delta V, \Delta A$) are Z-score normalized using the mean (μ) and standard deviation (σ) of the training set. This stabilizes training by centering the distribution and ensures that the loss function treats valence and arousal variations on a comparable scale.

5.1.3 Experimental Setup

Data Splitting Strategy. We implemented a *User-Specific Temporal Split*: for every user, the first 80% of chronological interactions were assigned to the training set, and the subsequent 20% were reserved for validation. Users with fewer than 6 events were assigned entirely to the training set to preserve sufficient history for context windowing.

Reproducibility. Hyperparameters were fixed across experiments to ensure reproducibility. The complete configuration is detailed in Table 3.

Parameter	Value
History Window (k)	4
Max Sequence Length	320 tokens
Batch Size	4
Epochs	16
Learning Rate	$1 \cdot 10^{-5}$
Weight Decay	0.01
Gradient Clipping	1.0
Loss Function	Huber ($\beta = 0.3$)
Seeds	{0, 100, 200}

Table 3: Hyperparameters used for the Subtask 2A forecasting experiments.

5.1.4 Results and Analysis

Discussion. Table 4 presents the results aggregated across the seeds abovementioned. Comparing the tested models against the SemEval baseline reveals several key findings:

- **Performance Overview:** *Twitter-RoBERTa* is the top-performing model, showing the highest improvement over the baseline for both Valence ($r +0.233$) and Arousal ($r +0.289$). It also achieves the lowest MAE, demonstrating robust predictive capability.

Table 4: Validation results for Subtask 2A compared with SemEval baseline. We report Pearson correlation (r) and MAE averaged across the random seeds above-mentioned.

Model	Valence (V)		Arousal (A)	
	r	MAE	r	MAE
SemEval Baseline linear(BERT)	.290	1.294	.199	0.744
Tested Models				
roberta-base	0.226	1.029	0.151	0.633
twitter-roberta-base-emotion	0.523	0.946	0.488	0.618
roberta-base-go_emotions	0.399	1.000	0.353	0.630
Difference with baseline				
roberta-base	-0.064	-0.265	-0.048	-0.111
twitter-roberta-base-emotion	+0.233	-0.348	+0.289	-0.126
roberta-base-go_emotions	+0.109	-0.294	+0.154	-0.114

- **Impact of Domain Shift:** While *RoBERTa-base* improves MAE compared to the baseline, its average Pearson correlation is lower than the linear baseline (-0.064 for Valence). This drop is attributed to high instability across seeds, where the model failed to converge in some initializations, dragging down the average performance. This confirms that generic pretraining is less reliable for this specific task.
- **Emotion vs. Social Pretraining:** *RoBERTa-GoEmotions* outperforms the baseline but lags behind *Twitter-RoBERTa*. This suggests that while emotion-specific fine-tuning helps, the broader adaptation to social media language provided by *Twitter-RoBERTa* is more beneficial for interpreting the informal "feeling words" found in the dataset.

5.2 Subtask 2B: Dispositional Change

While Subtask 2A focuses on immediate volatility, Subtask 2B addresses the prediction of broader shifts in a user’s emotional baseline. The objective is to forecast the difference between the average affect of a future period ($t+1$ to n) and the observed history (1 to t). The target variable Δ_{avg} is defined as:

$$\Delta_{avg} = \text{avg}(v_{t+1:n}) - \text{avg}(v_{1:t}) \quad (3)$$

5.2.1 Dataset Analysis

Subtask 2B relies on a user-level forecasting dataset derived from the longitudinal affect corpus, specifically structured to model long-term dispositional change. Each instance corresponds to a single user and contains two temporally disjoint segments: an *observed history* and an *equally-sized future segment*. The learning target is the difference between the average affect of these two segments.

Data Structure. The dataset is provided in a flattened, text-level format with the following key fields: `user_id`, `text`, `timestamp`, `group`, `valence`, and `arousal`. The group variable distinguishes between:

- **group==1:** texts belonging to the observed segment,
- **group==2:** texts belonging to the future segment (used only to compute gold labels).

For each user, the dataset additionally includes pre-computed target values: `disposition_change_valence` and `disposition_change_arousal`, corresponding to Δ_{avg} . This organization reflects a realistic forecasting scenario: while gold labels depend on future affect, no future textual content is available at test time.

User-Level Temporal Characteristics. Users exhibit substantial variability in the number of observed texts, temporal span, and sampling density. Many users provide only a small number of historical entries, while others contribute long and irregular sequences spanning multiple months or years. This heterogeneity introduces two major challenges:

- **Data sparsity:** limited history weakens estimation of stable baselines,
- **Irregular sampling:** large and inconsistent time gaps complicate trend estimation.

These properties motivate representing each user through aggregated statistics and trend-based descriptors rather than relying exclusively on sequential token-level modeling.

Distribution of Dispositional Change. The target variables (ΔV_{avg} , ΔA_{avg}) are approximately centered around zero, indicating that most users do not undergo extreme baseline shifts between observed and future windows. However, the distributions display noticeable variance and long tails, reflecting that a non-negligible subset of users experiences meaningful positive or negative affective drift. This characteristic supports the use of regression models with robust loss functions, as large deviations may correspond either to genuine psychological changes or to noisy self-reports.

5.2.2 Methodology

To capture long-term affective drift without accessing future information, we propose a **Multi-Modal Fusion** approach that integrates aggregated textual history with trajectory-based numeric features.

User-Level Input Construction. We construct a single instance per user using only entries from the observed segment (group=1) to strictly prevent data leakage.

- **Textual Stream:** The last $K = 6$ observed texts are sampled and concatenated using the separator token $\langle /s \rangle$, providing a compact summary of the most recent semantic context.
- **Numeric Trajectory:** We compute 10 scalar features derived solely from the observed timeline: mean and standard deviation of Valence and Arousal, total observation span (days), first-last deltas, and least-squares slopes over time.

Fusion Architecture. We employ mental-roberta-base as backbone due to its domain adaptation to mental-health-related discourse. The textual sequence is encoded by the Transformer and pooled from the last hidden states. Numeric features are projected through a small MLP and concatenated with the text embedding. A final regression head jointly predicts $(\Delta V_{avg}, \Delta A_{avg})$.

5.2.3 Experimental Setup

Hyperparameter	Value
Tokenization max length	256
Observed texts (K)	6 (Last-K strategy)
Batch size	8
Gradient accumulation steps	2
Epochs	50 (Early stopping: 8)
Optimizer	AdamW
Dropout	0.2
Seeds	{0, 50, 100, 200}

Table 5: Hyperparameters for Subtask 2B Fusion Model.

Validation Strategy. We use 4-fold **GroupKFold** cross-validation grouped by user_id, ensuring that users do not appear in both training and validation sets within a fold.

Preprocessing and Optimization. Targets and numeric features are Z-score normalized using training-fold statistics. We optimize the model

with Smooth L1 (Huber) loss ($\beta = 1.0$) with asymmetric weighting (Valence 1.5, Arousal 1.0). Discriminative fine-tuning is applied: the Transformer backbone is frozen except for the last three encoder layers, trained with a smaller learning rate (1×10^{-5}) than the regression head (5×10^{-4}). Training is performed with **early stopping** based on validation MAE, using a patience of 8 epochs to prevent overfitting.

5.2.4 Results and Analysis

Table 6 reports validation results averaged across four random seeds.

Model	Valence (V)		Arousal (A)	
	r	MAE	r	MAE
SemEval Baseline				
linear(prev)	.434	.406	.584	.286
Tested Models (Avg. seeds)				
roberta-base	.737	.555	.792	.306
twitter-roberta-base-emotion	.712	.608	.718	.333
roberta-base-go_emotions	.693	.660	.658	.388
mental-roberta-base	.745	.543	.793	.287
Difference with baseline				
roberta-base	+0.303	+0.149	+0.208	+0.020
twitter-roberta-base-emotion	+0.278	+0.202	+0.134	+0.047
roberta-base-go_emotions	+0.259	+0.254	+0.074	+0.102
mental-roberta-base	+0.311	+0.137	+0.209	+0.001

Table 6: Validation results for Subtask 2B compared with SemEval baseline (*linear(prev)*).

Discussion. All Transformer-based fusion models substantially outperform the SemEval *linear(prev)* baseline, confirming the importance of deep contextual representations for forecasting long-term affective drift. Among them, the proposed **mental-roberta-base** fusion model achieves the best overall performance, obtaining the highest correlation for both Valence and Arousal and the lowest MAE for Arousal.

Unlike Subtask 2A, where social-media-oriented pretraining was most beneficial, Subtask 2B favors representations learned from mental-health-related corpora. This suggests that dispositional change reflects slow, introspective emotional processes that are better captured by models exposed to longitudinal and psychologically grounded language rather than short, reactive expressions typical of social media.

6 Conclusion and Future Works

Our study demonstrates that effective longitudinal affect assessment requires task-specific architectural specialization. For **Subtask 1**, **RoBERTa-GoEmotions** proved superior, especially in arousal

prediction ($r = 0.528$). This success stems from integrating **User Embeddings** to model individual "affective baselines" and employing **Huber Loss** to stabilize training against ecological noise.

Regarding **Task 2**, we observed a clear performance dichotomy. Short-term state changes (**2A**) are best captured by **Twitter-RoBERTa**, highlighting the importance of informal linguistic priors and our context-aware metadata serialization. Conversely, long-term dispositional drift (**2B**) benefits from **Mental-RoBERTa**, where psychologically-grounded representations coupled with numeric trajectory features provide a stable estimate of affective drift.

Future work will focus on: (i) explicit temporal modeling via continuous-time embeddings to handle irregular gaps; (ii) dynamic personalization strategies beyond static embeddings; and (iii) uncertainty-aware frameworks to differentiate genuine emotional shifts from reporting bias. Finally, we emphasize the critical need for ethical oversight and privacy safeguards in deploying affect forecasting systems.

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